Impacts of Public Early Childhood Education in Chile on Maternal Fertility and Labor Market Outcomes

Short-run effects — progress update

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Where we are

What we do

Question: does opening a public preschool nearby change mothers' labor supply and fertility?

Variation: 319 neighborhood units (UVs) open their *first* JUNJI/Integra preschool in 2015–2024 vs. 4,174 that never do

Design: staggered event study at the UV level, Sun–Abraham estimator (Baker et al., *JEL* 2026)

Data: woman-level panel linking administrative registries — RSH/FPS, SII/AFP earnings, Civil Registry births, preschool enrollment

New: corrected opening years and UV crosswalk; vulnerable women; mothers followed until 2024

What we find so far

First stage: enrollment **+3.6pp** (34% → ≈38%), twice the previous estimate, from the opening year

Formal employment: gradual, **+0.9pp** on average in the first five years

Earnings incl. zeros: **+CLP 11k a month** in the first five years; half from more mothers working, half from higher earnings

Fertility: no effect — also with woman fixed effects

Project status

Done: corrected treatment, all outcomes re-estimated on the new samples

In progress: UV crosswalk review, woman FE labor results, pre-trend checks

Next: balanced panel and earlier years, mechanisms, writing

Research question

Question

What are the effects of the public expansion of early childhood education in Chile on mothers' labor market participation and fertility?

What we know

Mixed evidence: positive effects on maternal employment in Quebec and Argentina, close to zero in Norway

Recent Latin American evidence comes from **lotteries and experiments** (Rio de Janeiro, Nicaragua)

Less is known about a **national, decade-long public expansion** observed in administrative data

Our contribution

First study to use neighborhood-unit (UV) variation: preschools geolocated to the UV and linked to individual administrative records

Fertility as a second margin, alongside labor supply

What is a UV? *Unidad Vecinal*: the official sub-municipal territorial unit — ∼6,900 UVs vs. 346 municipalities.

⇒ **Finer geography: a sharper measure of exposure and more variation to identify the effect.**

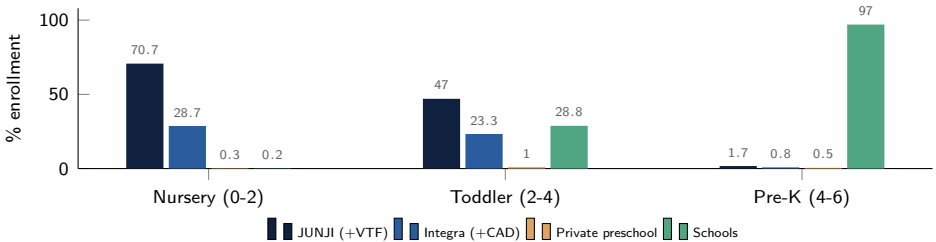
Quebec: Baker et al. (2008); Argentina: Berlinski & Galiani (2007); Norway: Havnes & Mogstad (2011); Rio: Attanasio et al. (2025); Nicaragua: Hojman & López Bóo (2022).

Public provision expanded sharply between 2015 and 2024

Mixed provision: **public** (JUNJI, Integra, municipal/SLEP) vs. **private** (subsidized/paid)

700+ new public preschools opened between 2015 and 2024

Net attendance rate (ages 0–5) rose from **50% to 55%** over the period, despite pandemic-era setbacks



Source: authors' elaboration based on Figura 5, Matrícula Oficial 2023, Centro de Estudios MINEDUC.

⇒ **Ages 0–4: JUNJI and Integra are the main providers — our focus.**

Identification: staggered-adoption event study

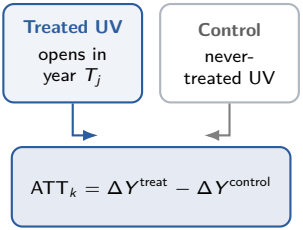
Design: staggered-adoption difference-in-differences with event study

Unit of treatment: the UV, the year it opens its first public preschool

Unit of analysis: **UV** $\times$ **year** panel, built from a woman $\times$ year panel linked to each woman's UV of residence

Fixed effects: **UV + year** (baseline); woman FE on the woman $\times$ year panel as a **robustness check**

Estimand: ATT of UV-level treatment (Baker et al., *JEL* 2026) — an **intent-to-treat effect of access**, not individual take-up



$\Rightarrow$ **We compare changes in UVs that get their first preschool vs. UVs that never do — an intent-to-treat effect of access.**

A UV is treated when it opens its first preschool

Treatment definition

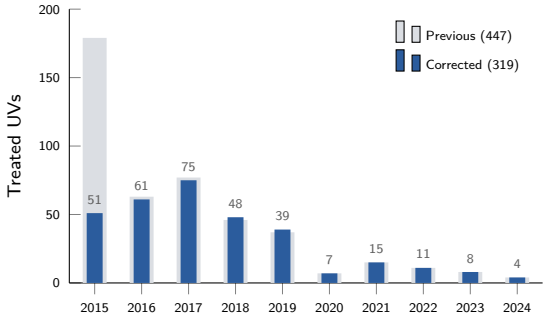
A UV is **treated** the year it opens its first public preschool (JUNJI/Integra) — goes from 0 to 1+ centers.

Group	Definition	N (UVs)	%
Excluded	Already had 1+ center in 2014 (out of design)	2,394	34.8%
Never-treated	0 in 2014, stays at 0 through 2024	4,174	60.6%
Treated	0 in 2014, opens first center 2015–2024	319	4.6%
Total universe		**6,887**	100%

Source: `base_cobertura_cp.dta`, corrected opening years (Oct 21). Design sample (`muestra_t1==1`) = 4,493 UVs. Previous version: 447 treated, 2,265 excluded.

⇒ **We identify the effect of the first preschool (0 → 1): 319 treated vs. 4,174 never-treated.**

• Treatment cohorts by year



2015 cohort corrected: 179 → **51 UVs** — in 127 UVs the "opening" was a center already operating in 2014 (next slide)

Other cohorts barely change; **2017 is now the largest** (75 UVs)

59% open in 2015–2017; only 45 UVs open after 2019

The 2015 cohort still has **only one pre-period** (2014)

Source: `base_cobertura_cp.dta`, corrected opening years (Oct 21). 319 treated UVs (0 centers in 2014).

⇒ **Openings are spread over 2015–19, no longer concentrated in 2015.**

• Resolved: most 2015 "openings" were centers already operating

Problem: JUNJI opening dates come from records that start in March 2015 → centers with no earlier record were dated **01-03-2015** (84% of 2015 openings)

Evidence: older transparency requests give **center** × **year panels** (JUNJI 1996–2023, Integra 2001–2024); JUNJI centers are flat at ∼3,150 a year in 2011–2016

Fix: opening year = first year the center appears in the panels

JUNJI centers "opening" in 2015

Recorded date	Centers	Already there
01-03-2015	444	**427 (96%)**
Other 2015 date	108	15 (14%)

UVs	Before	After
Treated	447	**319**
2015 cohort	179	**51**
Excluded (center in 2014)	2,265	2,394
Never-treated	4,175	4,174

Done: everything re-estimated with the corrected treatment

All results in the main slides use the corrected treatment (319 treated UVs). Previous results are in the appendix.

⇒ **The 2015 cohort was mostly a data artifact: 127 of its 179 UVs already had a center in 2014.**

Empirical specification and estimator

$$Y_{ut} = \lambda_u + \lambda_t + \gamma\, \overline{\text{Age}}_{ut} + \sum_{k \neq -1} \beta_k\, \mathbf{1}\{t - G_u = k\} + \varepsilon_{ut}$$

Y_{ut}: average outcome of women in UV u, year t (built from the woman $\times$ year panel). λ_u, λ_t: UV and year fixed effects; SE clustered by UV.

$\overline{\text{Age}}_{ut}$: average age of mothers in the UV-year (labor market specification).

β_k: effect k years from opening, relative to $k = -1$ (omitted); $k \leq -5$ binned.

Estimation strategy

Baseline TWFE as reference — but with staggered treatment, TWFE can use already-treated units as controls (Goodman-Bacon problem). Robust estimator: **Sun–Abraham** (`eventstudyinteract`), never-treated UVs as control group. Framework: Baker, Callaway, Cunningham, Goodman-Bacon & Sant'Anna (*JEL*, 2026).

Main threat: non-random placement of openings (JUNJI/Integra prioritizing UVs by care demand or socioeconomic profile). **UV FE** absorb permanent differences across UVs; **pre-period coefficients** ($k < -1$) test whether trends were already diverging.

$\Rightarrow$ **Main estimator: Sun–Abraham with never-treated UVs as controls; TWFE as reference.**

Data: what we have and how each sample is built

Sample	Source data	Key variables
Coverage (UV × year)	JUNJI/Integra transparency requests + UV shapefile (MDSF)	Number of centers, treatment status, opening cohort
Women panel (base)	RSH (2016–2024) + FPS (2014–2015)	UV of residence, UV population, child density
Labor market	+ Rentas (SII/AFC) + Cotizaciones (AFP)	Annual income, months worked, employment
Fertility	+ Civil Registry birth records (via RIS)	Birth in the year, children under 5
First stage (enrollment)	+ Matrícula parvularia (child-level)	Enrollment, enrollment rate

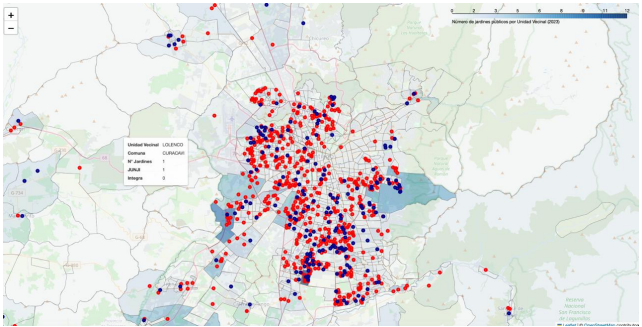
⇒ **All administrative registries, linked at the individual level — no survey self-reports.**

Every preschool is matched to its 2024 neighborhood unit

Each JUNJI/Integra preschool is **geolocated** and matched to the UV where it sits (spatial join)

UV boundaries **changed a lot** over time → we traced them and mapped everything to **2024 boundaries**

Preschools and women's UV of residence share **one common geography** across all years



Región Metropolitana. ● JUNJI ● Integra Shading: public preschools per UV. Source: `base_cobertura_cp.dta`.

⇒ **One common geography: every preschool and every woman mapped to 2024 UV boundaries.**

▸ Interactive map 2014–2024

• Who is in each sample, and what we measure

1. Samples (60% most vulnerable, fixed per woman)

First stage: children aged 0–4 of vulnerable mothers

Labor market: vulnerable mothers, **followed from the first year they have a child <5 until 2024**

Fertility: vulnerable women aged 18–49

2. Outcomes (averaged at UV × year)

First stage: enrollment rate

Labor market: share with formal earnings; earnings including zeros (relative and in CLP); earnings of working mothers

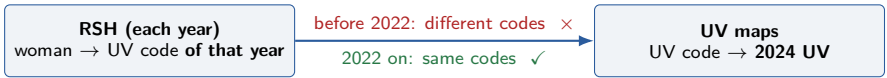
Fertility: share of women who had a child that year

Vulnerable = FPS score ≤13,500 (2014–2015) or RSH bracket ≤60 (2016 on), classification in the first year the woman appears. **Earnings** = taxable income + independent income.

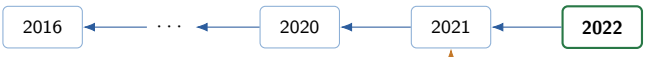
⇒ **All results today: vulnerable women, mothers followed over time, corrected treatment and UV crosswalk.**

• UV crosswalk: improved, still under review

The problem



The fix: follow the households, back from 2022



Example: code **A** in 2021 has 10 households. In 2022, **8** of them live in UV 105 and 2 in UV 107 → **A = UV 105**.

Before: some years were matched by code similarity and others by following households → a **break** between years.
Now: follow the households for **every year before 2022** → no break. Still reviewing it: the Ministry (MDS) does not have the official code equivalences.

⇒ **One consistent method for all years; we keep checking it.**

What we fixed in the data since the last meeting

1. Opening year of each preschool

The 2015 problem: fixed → treated UVs **447** → **319**

2. UV crosswalk

Revised: some women were assigned to the wrong 2024 UV (previous slide)

3. Labor samples and variables

Mothers stay in the sample until 2024

Earnings: before, **only taxable (dependent) income** if she had any; now **taxable + independent income**

Why this matters so much

Fixes 1 and 2 correct **who is treated**

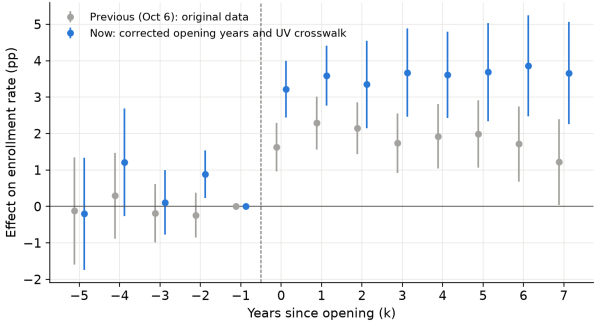
A UV wrongly counted as treated has no real change → it pulls the average effect **toward zero**

128 of the 447 "treated" UVs had no new center → their zero effect was averaged in and pulled the estimate down

Women placed in the wrong UV get the wrong treatment → same attenuation

⇒ **Fixing who is treated removes attenuation: effects should rise — and they do.**

• First stage: the effect doubles once the data are fixed



Y-axis: change in the enrollment rate of children 0–4, treated vs. never-treated UVs, relative to the year before opening

Starting levels (year before opening): treated 31%, control 33%

Now: **+3.6pp** on a 34% base (≈+10%), vs. +1.8pp before

Jumps at opening and stays flat

≈**5 more children** enrolled per UV

⇒ A new preschool raises enrollment from 34% to ≈38%: twice what we had before the fixes.

Children of vulnerable mothers. Sun–Abraham, weighted by number of children, 95% CI clustered by UV. Average k = 0–7. k = 8–9 omitted (only 2015–2016 cohorts).

• Why is the first stage still small?

What we already checked

✓ **Opening year** (2015 cohort): fixed → effect up
✓ **UV crosswalk**: fixed → effect up
✓ **Vulnerable mothers only**: barely changes it
✓ **Cleaner control group**: same → not spillovers
⇒ +1.8pp → **+3.6pp**, ≈5 children per UV

What our outcome measures

Whether children **living** in the treated UV are enrolled in **any** preschool: a **net** change

A new preschool can fill up without moving our outcome

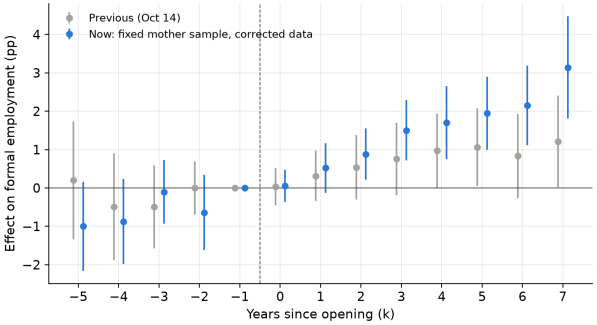
Switching: local children already enrolled elsewhere move to the new, closer preschool → no net change, but mothers save time

Mothers use other preschools, e.g. near work → new places go to children from other UVs

Next: enrollment records show which preschool each child attends → who fills the new preschool and where they were before.

⇒ **Net enrollment moves little, but switching to a closer preschool can still help mothers.**

• Mothers' formal employment rises gradually after the opening



Y-axis: change in the share of vulnerable mothers with formal earnings

Starting levels (year before opening): treated 52%, control 52%

Zero at opening, significant from $k = 2$

First five years ($k = 0$–4): **+0.9pp** on average (≈+2%) → **≈2 more working mothers** per UV (≈230 vulnerable mothers); over $k = 0$–7: +1.5pp

Counts *all* mothers in the UV, users or not (intent to treat)

⇒ **A new preschool gradually raises vulnerable mothers' formal employment: ≈2 more working mothers per UV in the first five years.**

Vulnerable mothers followed until 2024. Sun–Abraham, weighted, age controls, 95% CI clustered by UV.

• Employment: is it a pre-existing trend?

Pre-period coefficients are slightly negative; seen together, the plot looks like one rising line

But pre and post come from **different cohorts**: $k = -5$ only from 2019+ openings, $k = 5$–7 only from 2015–2017

2016 and 2017 (largest cohorts with pre-years): **no rising trend** before opening (2017 was even falling); after opening they rise ≈3 points, gradually

The negative pre-coefficients come from the **2020–2024 cohorts**: small and noisy

Within each cohort (raw gap, treated − control, pp)

Cohort	$k = -2$	-1	0	2	4	7
2015		−3.8	−3.4	−1.3	−1.7	−0.6
2016	−3.0	−3.6	−2.8	−0.6	−1.0	−0.2
2017	−1.1	−1.4	−1.9	−1.1	−0.3	+1.1

Still missing: joint pre-test and HonestDiD (covariance matrix in next extraction); event study balanced in event time.

⇒ **In the cohorts we follow longest, employment was not trending up: the rise starts after the opening.**

• Earnings levels: half of vulnerable mothers have no formal earnings

			Monthly earnings (CLP of 2024)	
Year	Mothers	With earnings	All (incl. zeros)	Working mothers
2014	242,897	50%	241k	481k
2016	333,493	52%	274k	530k
2018	394,208	52%	299k	574k
2020	457,404	49%	308k	627k
2022	541,063	57%	377k	663k
2024	565,324	56%	429k	769k

The sample grows every year: new mothers enter and none leave

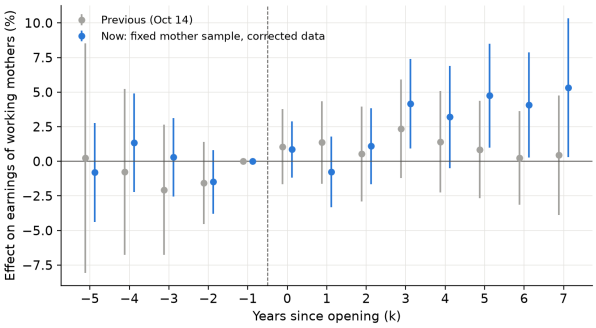
Share with earnings rises from 50% to 56%, with a dip in 2020 (COVID)

Annual earnings / 12: mothers who worked part of the year show less

⇒ **Half of vulnerable mothers have no formal earnings in a given year; those who work earn ≈CLP 770k a month (2024).**

Vulnerable mothers followed until 2024, all UVs. Median for working mothers in 2024: CLP 631k. Deflated with the CPI (annual averages, approximate).

• Working mothers now also earn more



Y-axis: % change in earnings of mothers **who work**

Starting levels (2014): treated and control earn the same

First five years ($k = 0$–4): **+1.7%** on average (≈CLP 14k on CLP 807k); clearer from $k = 3$; over $k = 0$–7: +2.8%

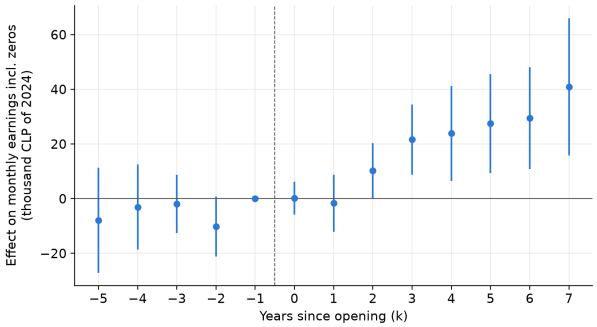
Before: ≈0 — now earnings include independent income and mothers are followed longer

New workers probably earn less → the gain for those already working may be larger

⇒ **Mothers who already worked also earn somewhat more after the opening (≈2% in the first five years).**

Only mothers with earnings. Sun–Abraham, weighted by number of mothers, 95% CI clustered by UV.

• Earnings including zeros rise gradually after the opening



Y-axis: monthly earnings per mother, counting **zero** for those without earnings (CLP of 2024); control mean CLP 427k

Starting levels (2014): similar, treated ≈2% above control

First five years ($k = 0$–4): **+CLP 11k** (≈+2.5%, SE 4k), significant from $k = 2$; over $k = 0$–7: +CLP 19k (SE 6k)

Same gradual shape as employment

No pre-trends (joint test does not reject)

⇒ Mothers in the UV earn ≈CLP 11k more a month in the first five years after the opening.

Earnings relative to the year's mean, rescaled to the 2024 mean (relative version: Appendix A14). Before: +2% (Oct 14).

Sun–Abraham, weighted by number of mothers, 95% CI clustered by UV.

• Where the earnings gain comes from

Average earnings = (% working) × (earnings of those working): 427k = 0.53 × 807k

Channel	CLP/month
More mothers working: +0.9pp × 807k	7.5k
Higher earnings: 0.53 × 1.7% × 807k	7.3k
Sum / direct estimate	15k / 11k

First five years ($k = 0$–4). New workers probably earn less than 807k → the first channel is overstated.

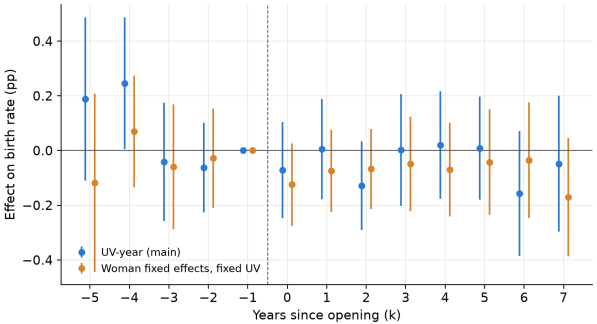
Why would working mothers earn more?

Our first stage is **net** enrollment (≈5 children). Many mothers probably **switch** their child from a farther preschool to the new, closer one: no change in enrollment, but less commuting → more hours or better jobs.

To check: children in the new preschool who were enrolled elsewhere the year before.

⇒ **About half of the gain comes from more mothers working and half from higher earnings — consistent with switching to a closer preschool.**

• No effects on fertility yet



Y-axis: change in the share of vulnerable women 18–49 who had a child (pp)

Blue: UV-year, as in the labor results

Orange: woman fixed effects — each woman compared with herself, treated by her first UV

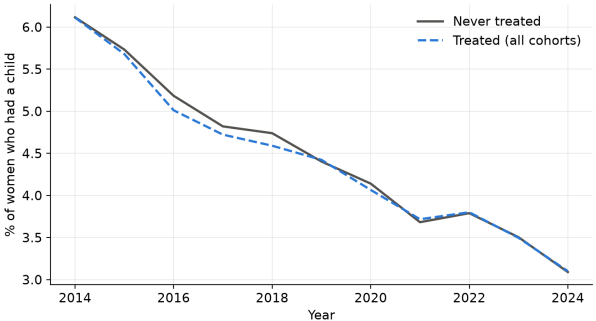
First five years ($k = 0$–4): $-$**0.03pp** (SE 0.07) on a 4.4% base; woman FE: -0.08pp $\rightarrow$ **both zero**

With woman FE the early pre-period bumps disappear

$\Rightarrow$ **Two different estimates give the same answer: no effect on fertility so far.**

Sun–Abraham, weighted, 95% CI clustered by UV. Woman FE: 12.4M woman-years.

• Fertility levels: treated and control fall together



Birth rate of vulnerable women 18–49 **halves**: from ≈6 to ≈3 births per 100 women a year (Chile's fertility decline)

Treated and control UVs fall **exactly together**, before and after the openings

Starting levels (2014): 6.1% in both

⇒ **Fertility in UVs with a new preschool moves exactly like in UVs without one.**

Share of women aged 18–49 who had a child in the year. Treated = all cohorts pooled.

• Woman fixed effects: what we built

Design

Each woman compared **with herself** before and after the opening

Fixed UV: her first observed UV, kept if she moves (intent to treat) → moves do not change who is treated

Balanced panel of women: enter in 2014, observed all 11 years → treated and control at the same stage of motherhood

Balanced panel check

≈79% of 2014 mothers stay all years, in control and in almost every treated cohort

Results

Fertility: zero, same as UV-year (previous slides)

Labor: estimated in the RIS, extraction failed → next week

⇒ **Individual-level estimates confirm the fertility zero; labor results arrive next week.**

Fertility: vulnerable women 18–49, 12.4M woman-years. Balanced share: share of mothers in the 2014 sample observed 2014–2024.

Woman fixed effects: specification

$$E_{it} \;=\; \mu_i \;+\; \lambda_t \;+\; \sum_{k \neq -1} \beta_k \, \mathbf{1}\{\, t - G_{u_0(i)} = k \,\} \;+\; \varepsilon_{it}$$

```
eventstudyinteract empleo_formal d1-d4 d6-d15, cohort(g1_f) control_cohort(nunca_f)
    absorb(rut_inn anio) vce(cluster uv_fija)
```

E_{it}: formal employment of mother i in year t

μ_i: **woman** fixed effect; λ_t: year fixed effect

$u_0(i)$: **fixed UV** = first UV where the mother appears, before the opening

$G_{u_0(i)}$: year her fixed UV opens its first preschool $\rightarrow$ kept even if she moves

Sun–Abraham: cohorts by $G_{u_0(i)}$ (`g1_f`); controls = fixed UV never opens one (`nunca_f`)

$k \leq -5$ binned; ref. $k = -1$; SE clustered by fixed UV

Sample: vulnerable mothers whose fixed UV had no preschool in 2014, first observed before the opening

$\Rightarrow$ **Each mother is compared with herself, treated by the UV where she lived before the opening.**

Preliminary run: ≈5M mother-years, 1,634 clusters; estimated on an earlier version of the new mother sample, to be re-estimated.

- **To discuss: what do we mean by a "balanced panel"?**

Balanced in...	What it means	Where we are
Event time	Only cohorts observed in the whole window (e.g. $k = -3$ to $5 \rightarrow$ 2017–2019 openings): the same UVs pre and post	Tried in the first stage: only $\sim$200 UVs, very noisy
Women	The same women observed every year 2014–2024	Built for woman FE: $\approx$79% of 2014 mothers, equal in treated and control
UVs	Every UV observed every year	Not imposed: some UVs only appear in some years

Which of these do you have in mind? They answer different concerns: **changing cohort composition across** k (event time) vs. **changing sample of women** (women).

More pre-years would make any of them less costly $\rightarrow$ next slide

$\Rightarrow$ **We would like your input on which balanced panel you expect to see.**

• More pre-trend years: how far back can we go?

Source	Outcome	From
FPS (women, household)	sample, UV	2007
Preschool enrollment	first stage	2011
Earnings (SII, historical)	labor	2002
Contributions (AFP)	labor	2015
Civil Registry births	fertility	no limit

The catch: UVs before 2016

FPS has **no reliable UV** code

Our crosswalk only reaches back to 2016

Pre-2016 years must be anchored to the woman's **2016 UV** → women who left before 2016 are lost, and earlier moves are invisible

Proposal: start in **2011** → 3 more pre-years for every cohort (the 2015 cohort goes from 1 to 4). Labor 2011–2013: earnings only.

⇒ **Data exist back to 2011 for all outcomes, but UVs before 2016 have to be proxied by the 2016 UV.**

• Follow-up on your requests

Meeting	Request	Status
Oct 8	60% most vulnerable	**Done**
Oct 8	Follow mothers over time (fixed sample)	**Done**
Oct 8	Individual-level estimates	**Partly**: fertility done; labor next week
Oct 8	How many women move across UVs	**Done** → led to the crosswalk revision
Oct 8	Treatment by initial stock of preschools	**Not started**
Oct 8	Sample by predicted probability of pregnancy	**Not started**
Oct 15	2015 openings	**Done**: data error, fixed
Oct 15	Moves in 2022	**Done**: crosswalk artifact, fixed
Oct 15	Effects by child's age at opening	**Estimated**, under review
Oct 15	Spillovers to neighboring UVs (fertility)	**Code ready**, not run yet
Oct 15	First stage for the other 40%	**Pending**
Oct 15	Magnitudes in CLP	**Done**
Oct 15	Balanced panel of women	**Built**; results next week
Oct 15	Who fills the new preschool; extended hours	**Pending**
Oct 22	Re-estimate with the corrected coverage data	**Done**
Oct 22	Earlier years for pre-trends	**Availability checked**; to discuss
Oct 22	Include UVs with preschools in 2014 (Callaway–Sant'Anna)	**Not started**
Oct 22	All women, not only mothers (labor)	**Running**

⇒ **Most requests from October are done or running; the treatment and sample redefinitions are still to start.**

• Running and next steps

Running / next extraction

Labor outcomes for all vulnerable women 18–49, not only mothers

Woman FE labor results (estimated)

Employment covariance matrix → joint pre-test and HonestDiD

Data

Keep reviewing the UV crosswalk

Earlier years (2011–2013), after agreeing on the balanced panel

Pending

Event study balanced in event time

Sensitivity to pre-trends (HonestDiD): first stage, employment, earnings

Who fills the new preschool: test the switching hypothesis

Include UVs with preschools in 2014 (Callaway–Sant'Anna)

Treatment by initial stock; sample by probability of pregnancy

The other 40%: first stage and labor

Spillovers to neighbors (fertility)

Effects by child's age at opening